

LoRa TinyML Meter Reader

Automated meter reading via TinyML + LoRa for low-power, long-range IoT monitoring

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Introduction

- **Electricity metering remains a critical part of utility infrastructure, yet many systems rely on outdated manual processes**
- **With the rise of IoT and edge AI, there is an opportunity to modernize meter readings using low power autonomous technologies**

Problem Statement

Manual Meter Reading

Utility workers visit each meter individually. This is costly, slow, and error-prone across large geographic areas.

No WiFi in Remote Areas

Rural and off-grid locations lack reliable internet infrastructure, making cloud-connected solutions impractical.

High Power Consumption

Traditional IoT solutions (WiFi, cellular) are too power-hungry for battery-operated field deployments.

Literature Review

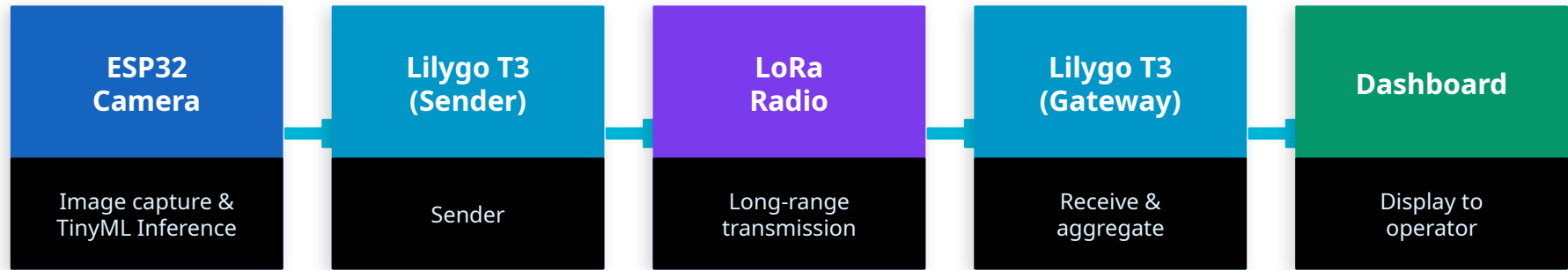
- Many papers created lightweight CNN based meter reading models [1] [2]
- Few actually turned it into a tinyML, and even fewer actually deployed it in the real world [3]
- None of the papers incorporate LoRa-based communications in their systems

Proposed Solution

- An automatic low power system that reads electricity meters on device AI and transmits results over LoRa
- Eliminates manual labor and network dependency while enabling long range energy efficient monitoring



System Architecture



Hardware Components

2× Lilygo T3 (LoRa + ESP32-S3)

1× ESP32 Camera Module

Wired connection: Camera → Sender unit

Implementation Plan & Timeline

Week 1	Week 2	Week 3	Week 4
Data & Comms Setup	Model Optimization	On-Device Deployment	Full Integration & Demo
• Send photos over LoRa • Capture dataset of meter images • Create/fine-tune base ML model	• Optimize model architecture • Convert to TensorFlow Lite Micro • Quantize weights, validate accuracy	• Deploy model to Lilygo T3 • Test inference accuracy on device • Tune performance vs. accuracy	• Integrate all subsystems • Build live dashboard • Measure power consumption
Dataset of meter images	Trained & quantized model	Deployable on-device model	Full working demo

Progress So Far

- We have an machine learning model that works on our PC, yet to be shrank
- A small internal dataset to test on



Challenges & Next Steps

Challenges

Model Size: MCU has limited flash/RAM, The ESP32 only has 520 KB SRAM and 4 MB PSRAM while the LilyGo has only 520 KB SRAM

Lighting Variation: Meter readability depends on ambient light; preprocessing needed.

Next Steps

Train & Quantize Model: Optimize our model even further to allow deployment, turning ~8 MB model to less than 300 KBs

High Accuracy: Get a high accuracy in the tinyML model

References

- [1] Sharma & Kim (2021), Lightweight CNN for Meter Digit Recognition, JSST
- [2] Saetan & Dittakan, “Thailand Water Meter Reading Using CNNs From Smartphone Imagery,” ECTI DAMT & NCON, 2024.
- [3] Nguyen et al. (2024), Real-Time Water Meter Reading Based on TinyML, EECSI

Thank You

Questions & Discussion

LoRa TinyML Meter Reader | IoT + Edge ML Project

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